

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: CSA1408

COURSE NAME: Compiler Design

COURSE OUTCOME COVERED

CO4: Examine intermediate code generation techniques to enhance code optimization (BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5

Duration :60 Mins
On Class
Total Marks:50

Annexure A

INDUSTRY PROBLEM- SOLVING TASK

Code of Conduct:

I Ranjitha R(192421285) _ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Q. No	Problem Understanding(20)	Method Selection(20)	Solution Process(25)	Accuracy of Calculation (20)	Interpretation of Results(15)	Total Score (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	
4	/ 4	/ 4	/ 4	/ 4	/ 4	
5	/ 4	/ 4	/ 4	/ 4	/ 4	
OVERALL RUBRICS VALUE						
	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 25	/ 25	/ 20	/ 20	/ 10	/ 100

Annexure A

INDUSTRY PROBLEM- SOLVING TASK

Q1. Smart Electricity Billing System

Concepts Covered: Intermediate Languages, Assignment Statements, Optimization

Problem Statement

An electricity billing system calculates the total payable amount using units consumed, rate per unit, and tax.

$$\text{energy_cost} = \text{units} \times \text{rate}$$
$$\text{tax_amount} = \text{energy_cost} \times \text{tax_rate}$$
$$\text{total_bill} = \text{energy_cost} + \text{tax_amount}$$

Tasks

- (i) Write suitable declarations for all variables with appropriate data types.
- (ii) Generate three-address intermediate code for the billing calculation.
- (iii) Represent the intermediate code using quadruples, triples, and indirect triples.
- (iv) Identify common subexpressions and optimization opportunities.
- (v) Produce the optimized intermediate code with justification. (10 Marks)

Q2. Employee Promotion Evaluation System

Concepts Covered: Boolean Expressions, Backpatching

Problem Statement

An employee is eligible for promotion if:

Experience $\geq$ 5 years

AND (Performance Score > 90 OR Certification = TRUE)

Tasks

- (i) Formulate the Boolean expression.
- (ii) Generate intermediate code using short-circuit evaluation.
- (iii) Apply backpatching to resolve forward jumps.
- (iv) Generate the final labelled intermediate code.
- (v) Draw the control-flow graph for the evaluation process. (10 Marks)

Q3. Hospital Patient Priority System

Concepts Covered: Case Statements, Intermediate Languages

Problem Statement

A hospital assigns treatment priority based on emergency level.

```
switch(priority)
```

```
{
```

```
case 1: CRITICAL;
```

```
case 2: HIGH;
```

```
case 3: MEDIUM;
```

```
case 4: LOW;
```

```
}
```

Tasks

- (i) Convert the switch statement into three-address code using conditional jumps.
- (ii) Generate an equivalent jump-table implementation.
- (iii) Represent the implementation using quadruples.
- (iv) Compare conditional branching with jump-table execution.

- (v) Recommend the most suitable implementation for emergency hospitals with justification.
(10 Marks)

Q4. E-Commerce Payment Processing System

Concepts Covered: Procedure Calls

Problem Statement

An online shopping application processes payments through the following procedures.

amount = calculateAmount(cart)

processPayment(amount)

sendConfirmation(amount)

Tasks

- (i) Generate intermediate code.
- (ii) Illustrate parameter passing.
- (iii) Design the activation record.
- (iv) Draw stack diagrams for call and return operations.
- (v) Explain how procedure calls are managed during execution. (10 Marks)

Q5. Smart ATM Withdrawal Verification System

Concepts Covered: Boolean Expressions, Backpatching

Problem Statement

A customer can withdraw cash if:

Account Balance $\geq$ Withdrawal Amount

AND (PIN Verified = TRUE OR Biometric Verified = TRUE)

Tasks

- (i) Write the Boolean expression.

- (ii) Generate short-circuit intermediate code.
- (iii) Apply backpatching.
- (iv) Generate the final labelled intermediate code.
- (v) Draw the control-flow graph. (10 Marks)

RUBRICS FOR CALCULATION

Numerical Problems					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Co-4

ASSESSMENT-2

NAME: RANJITHA R

REG. NO: 192421285

COURSE CODE: CSA1408

COURSE NAME: COMPILER DESIGN



Q1. Smart Electricity Billing system:

Given:

$$\text{energy_cost} = \text{unit} \times \text{rate}$$

$$\text{tax_amount} = \text{energy_cost} \times \text{tax_rate}$$

$$\text{total_bill} = \text{energy_cost} + \text{tax_amount}$$

i) Declaration:-

float unit;

float rate;

float tax_rate;

float energy_cost;

float tax_amount;

float total_bill;

ii) Three ADD Code:-

$$T_1 = \text{unit} * \text{rate}$$

$$T_2 = T_1 * \text{tax_rate}$$

$$T_3 = T_1 + T_2$$

$$\text{energy_cost} = T_1$$

$$\text{tax_amount} = T_2$$

$$\text{total_bill} = T_3$$

iii) Quadruples:-

No	operation	Arg 1	Arg 2	Result
1	*	units	rate	T_1
2	*	T_1	tax-rate	T_2
3	+	T_1	T_2	T_3
4	=	T_1	-	energy-cost
5	=	T_2	-	tax-amount
6	=	T_3	-	Total-bill

Triples :-

No	operation	Arg 1	Arg 2
1	*	units	rate
2	*	(0)	tax-rate
3	+	(0)	(1)
4	=	(0)	energy-cost
5	=	(1)	tax-amount
6	=	(2)	total-bill

Common subexpression/optimization :

There is no repeated subexpression in the gn calculation

$$T_1 = \text{units} * \text{rate}$$

$$\left. \begin{array}{l} \text{tax-amount} = T_1 * \text{tax-rate} \\ \text{total bill} = T_1 + T_2 \end{array} \right\} \begin{array}{l} \text{unnecessary} \\ \text{re-evaluation avoided.} \end{array}$$

i) optimized Intermediate Code :-

$$T_1 = \text{units} * \text{rate}$$

$$T_2 = T_1 * \text{tax-rate}$$

$$T_3 = T_1 + T_2$$

$$\text{energy-cost} = T_1$$

$$\text{tax-amount} = T_2$$

$$\text{total-bill} = T_3$$

Q2. Employee Promotion Evaluation system:-

i) Boolean Expression :-

Experience ≥ 5 AND (performance > 90 OR
certification == TRUE)

Let,

$$E = \text{Experience} \geq 5$$

$$P = \text{performance} > 90$$

$$C = \text{certification} == \text{TRUE}$$

Therefore,

$$E \text{ AND } (P \text{ OR } C)$$

ii) Short - Circuit Intermediate Code

if Experience < 5 goto L false

if performance > 90 goto L True

if certification == True goto L true

goto Lfalse

Ltrue :

Promotion = TRUE

goto Lend

Lfalse:

Promotion = FALSE

Lend:

iii) Back patching :-

Experience < 5 → ?

Performance > 90 → ?

Certification == TRUE → ?

After back patching :-

Experience < 5 → Lfalse

Performance > 90 → Ltrue

Certification == TRUE → Ltrue.

iv) Final Labelled Intermediate Code:

L1 : if Experience < 5 goto L4

L2 : if performance > 90 goto L3

L3 : if certification == TRUE goto L5

goto L4

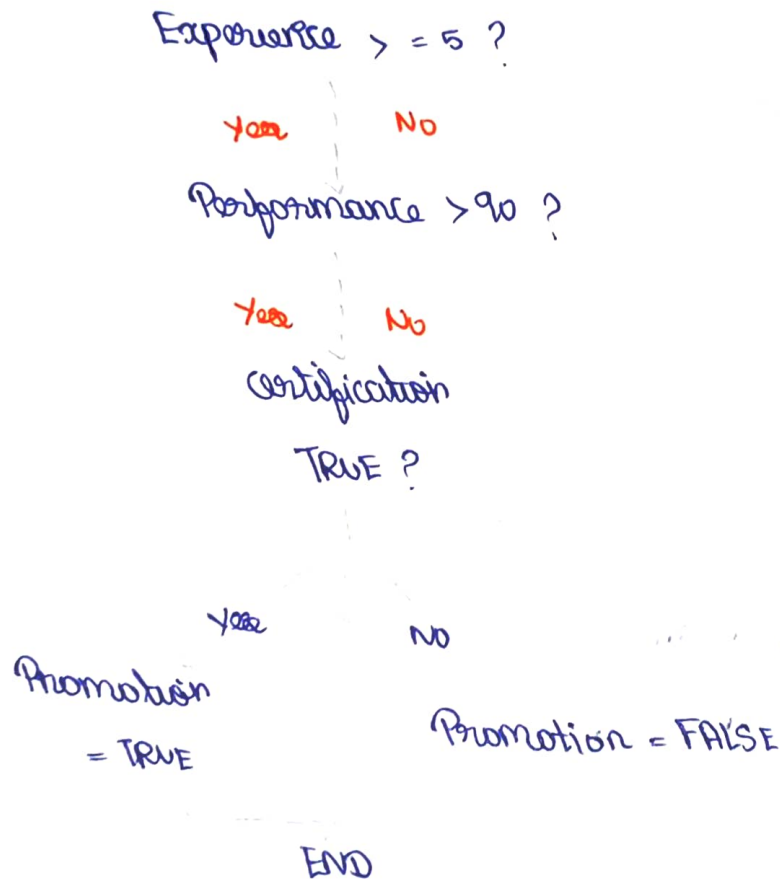
L5 : promotion = TRUE

goto L6

L4 : Promotion = FALSE

L6 : End.

✓ Control-Flow graph:-



Q3. Hospital patient Priority system:-

i) Three Address code using Conditional Jumps:-

if priority == 1 goto L1

if priority == 2 goto L2

if priority == 3 goto L3

if priority == 4 goto L4

goto L₅

L₁ : CRITICAL

goto L₅

L₂ : HIGH

goto L₅

L₃ : MEDIUM

goto L₅

L₄ : low

goto L₅

L₅ : END

ii) Equivalent Jump-Table Implementation :-

TT = priority - 1

goto Table [TT]

Jump table:

table [0] → L₁

table [1] → L₂

table [2] → L₃

table [3] → L₄

Examples:-

No	operation	Arg 1	Arg 2	Result:
1	= =	Priority	1	L ₁
2	= =	Priority	2	L ₂
3	= =	Priority	3	L ₃
4	= =	Priority	4	L ₄
5	-	Priority	1	T ₁
6	goto	Table [T ₁]	-	-

iv) Comparison:-

Conditional Branching

Jump Table.

uses multiple comparison

uses index directly

simple to implement

Faster for many cases

suitable for few cases

suitable for many consecutive cases.

more comparison may be required

Fewer comparison

v) Recommendation:-

Jump Table implementation is recommended for an emergency hospital system where there are many fixed and consecutive priority levels.

Q4. E-Commerce payment processing system:

The application performs:

amount = calculate amount (Card)

Process payment (amount)

Send confirmation (amount)

i) Intermediate Code:

Per ram Cost

T_1 = call calculateAmount, 1

Per ram T_1

call process payment, 1

Per ram T_2

call sendConfirmation, 1

amount = T_1

ii) parameter passing :-

call 1 $\rightarrow$ param Cost

call calculate Amount

amount = calculate Amount (Cost)

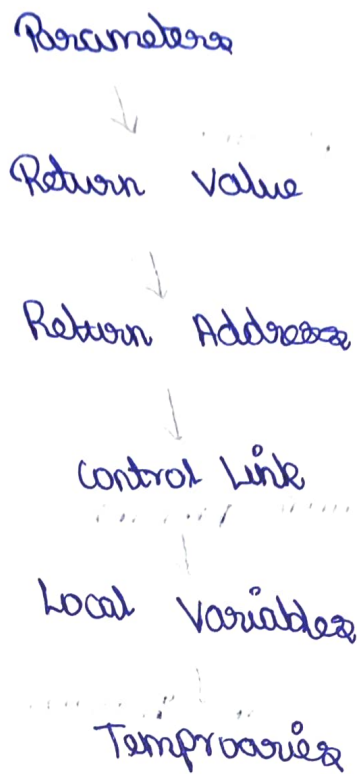
call 2 $\rightarrow$ param amount

call process payment

call 3 $\rightarrow$ param amount

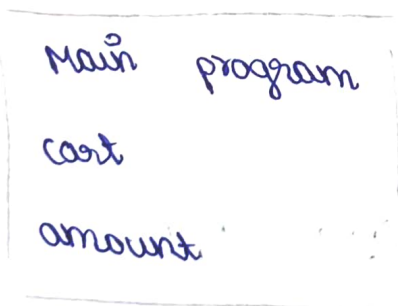
call send Confirmation:

iii) Activation Record:-

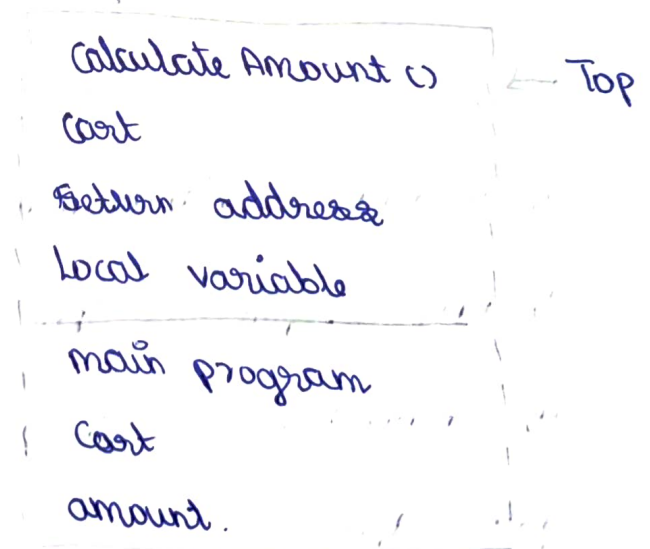


iv) stack Diagram:-

Before call



During calculate Amount()



v) procedure Call Management:-

→ Parameters are passed to the procedure

- An activation record created
- Return address is stored
- Procedure executes
- Return value is produced

Ex. Smart ATM withdrawal Verification system.

i) Boolean Expression:

$\text{Balance} \geq \text{Withdrawal Amount}$

AND

$(\text{PIN-Verified} == \text{TRUE} \text{ OR } \text{Biometric-Verified} == \text{TRUE})$

$B = \text{Balance} \geq \text{Withdrawal Amount}$

$P = \text{PIN-Verified} == \text{TRUE}$

$Bb = \text{Biometric-Verified} == \text{TRUE}$

ii) Short Circuit Intermediate Code

if $\text{Balance} < \text{Withdrawal Amount}$ goto Lfalse

if $\text{PIN-Verified} == \text{TRUE}$ goto Ltrue

if $\text{Biometric-Verified} == \text{TRUE}$ goto Ltrue

goto Lfalse

Ltrue :

$\text{Dispense-Cash} = \text{TRUE}$

goto Lend

Lfalse :

Dispense - Cash = FALSE

End :

ii) Back patching :-

Balance < withdrawal Amount $\rightarrow$?

PIN - Verified == TRUE $\rightarrow$?

Biometric - Verified == TRUE $\rightarrow$?

Balance < withdrawal Amount $\rightarrow$ Lfalse

PIN - Verified == TRUE $\rightarrow$ Ltrue

Biometric - Verified == TRUE $\rightarrow$ Ltrue.

iii) Final Labelled Intermediate Code :-

L1 : if Balance < withdrawal Amount goto L3

L2 : if PIN - Verified == TRUE goto L4

L3 : if Biometric - Verified == TRUE goto L4
goto L5

L4 : Dispense - Cash = TRUE

goto L6

L5 : Dispense - Cash = FALSE

L6 : END

Control Flow Graph:

Balance $\geq$ Withdrawal ?

Yes

No

REJECT

Pin Verified ?

Yes

No

Bio-Verified ?

Yes

No

REJECT

Dispense Cash